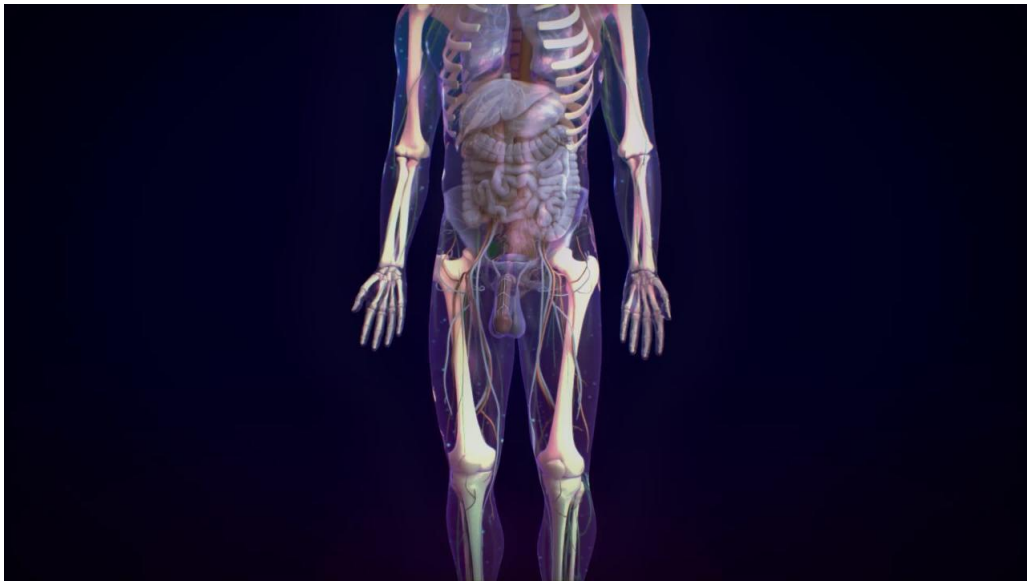






Erectile Dysfunction (ED)

(Impotence)

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Erectile dysfunction (ED) is the inability to attain or sustain an erection satisfactory for sexual intercourse.

(See also [Overview of Sexual Function and Dysfunction in Men.](#))

Every man occasionally has a problem achieving an erection, and such occurrences are considered normal. Erectile dysfunction (ED) occurs when a man

- Never achieves an erection
- Repeatedly achieves erection briefly but not long enough for intercourse
- Achieves effective erection inconsistently

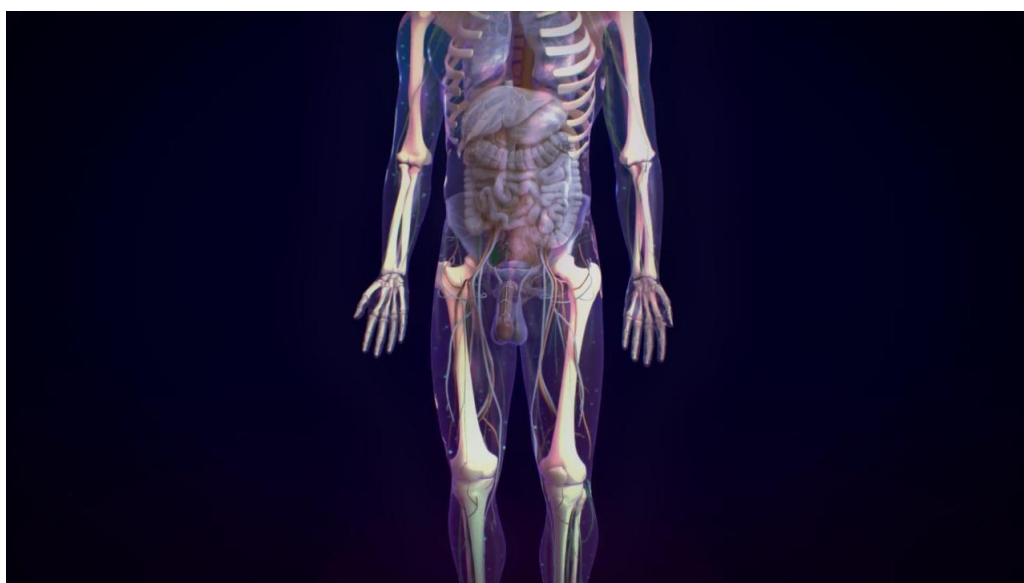
ED is called primary if the man has never been able to attain or sustain an erection.

ED is called secondary if it is acquired later in life by a man who was previously able to attain erections.

Secondary ED is much more common than primary ED.

In the United States, ED is very common and increases in prevalence with age, with the majority of men over the age of 50 being affected. However, ED can be successfully treated at any age.

Drug Information for the Topic



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Causes of Erectile Dysfunction

To achieve an erection, the penis needs an adequate amount of blood flowing in, a slowing of blood flowing out, proper function of nerves leading to and from the penis, adequate amounts of the male sex hormone testosterone, and sufficient sex drive (libido). A disorder of any of these systems may lead to erectile dysfunction (ED).

Most cases of ED are caused by abnormalities of the blood vessels or nerves of the penis. Other possible causes include hormonal disorders, structural disorders of the penis, use of certain medications, and psychological problems (see table

Common Causes and Features of Erectile Dysfunction). The most common specific causes are

- Hardening of the arteries (atherosclerosis) that affects the arteries to the penis
- [Diabetes mellitus](#)
- Complications of [prostate surgery](#) (such as robotic prostatectomy for prostate cancer)
- Certain medications, such as those used to treat high blood pressure or an [enlarged prostate](#) and those that act on the central nervous system, such as medications used to treat depression

Did You Know...

- Occasional inability to achieve an erection is normal and does not mean that a man has erectile dysfunction.

Often several factors contribute to ED. For example, a man with a slight decrease in erectile function caused by diabetes or peripheral vascular disease can develop severe ED after starting a new medication or if stress increases.

Blood vessel disorders

Atherosclerosis may partially block blood flow to the legs (peripheral vascular disease). Usually, arteries to the penis are also blocked, decreasing the amount of blood flow to the penis and causing ED. Diabetes, high cholesterol levels, high blood pressure, and smoking contribute to atherosclerosis and therefore to ED.

Sometimes blood leaks out of the veins in the penis too quickly, decreasing blood pressure in the penis and thus interfering with achieving or maintaining an erection (called veno-occlusive dysfunction or venous leak).

Nerve disorders

If the nerves sending messages to the penis are damaged, ED can occur. In addition to causing atherosclerosis, [diabetes](#) can also affect the nerves that supply the penis. Because nerves to the penis run along the prostate gland, prostate surgery (such as for cancer or an enlarged prostate) often causes ED.

Less common nerve disorders that cause ED include spinal cord injury, multiple sclerosis, and stroke. Also, prolonged pressure on the nerves in the buttocks and genital area (the so-called saddle area), as may occur during long-distance bicycle riding, can cause temporary or even permanent ED.

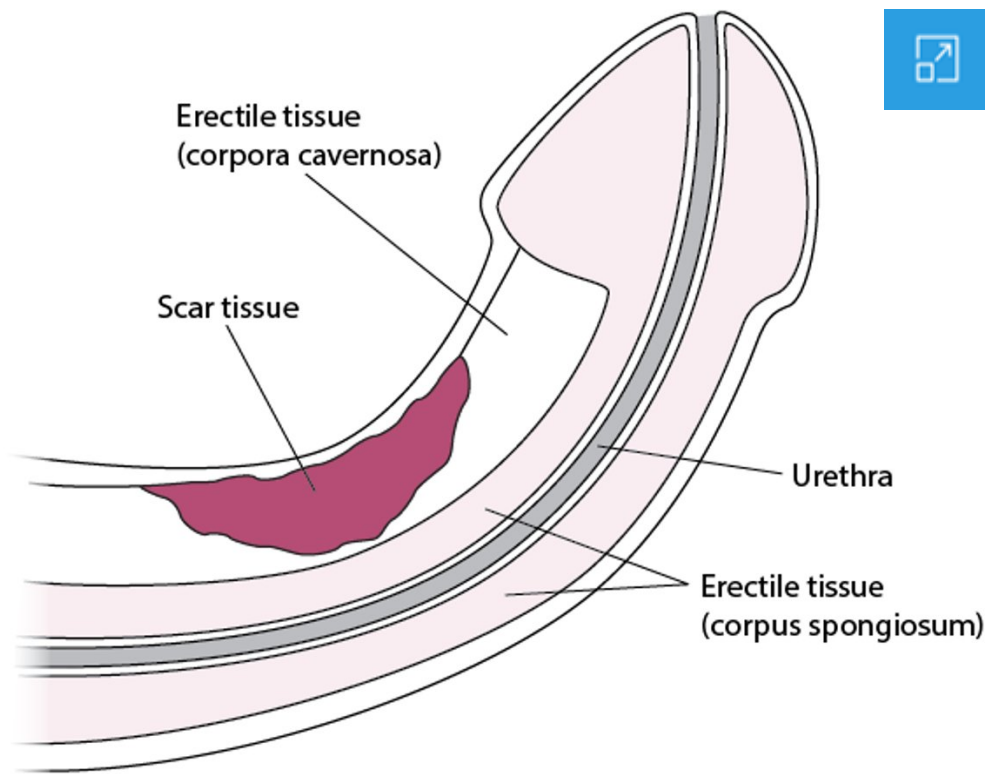
Other disorders

Hormonal disturbances (such as abnormally low levels of testosterone) tend to decrease sex drive but can also result in ED.

In [Peyronie disease](#), scar tissue develops inside the penis, resulting in curved and often painful erections and causing ED.

What Is Peyronie Disease?

In Peyronie disease, inflammation inside the penis causes scar tissue to form. Because the scar tissue does not enlarge during an erection, the erect penis is curved, making penetration during sexual intercourse difficult or impossible. The scar tissue may extend into the corpora cavernosa, causing erectile dysfunction.



Drugs, including alcohol and illicit drugs such as cocaine and amphetamines, can also cause or contribute to ED.

Sometimes **psychological problems** (such as performance anxiety or depression) or factors that decrease a man's energy level (such as illness, fatigue, or stress) cause or contribute to ED. Erectile dysfunction may be situational, involving a particular place, time, or partner.

Prolonged, painful erection (priapism) may damage the erectile tissue of the penis, leading to ED.

Evaluation of Erectile Dysfunction

An occasional episode of erectile dysfunction (ED) is not uncommon, but men who are consistently unable to achieve or maintain an erection should see their doctor because ED may be a sign of a serious health problem, such as [atherosclerosis](#) or a nerve disorder. Most causes of ED are treatable. The following information can help men know when to see a doctor and what to expect during the evaluation.

Warning signs

In men with ED, certain symptoms and characteristics are cause for concern. They include

- Absence of erections during the night or upon awakening in the morning
- Numbness in the area between and around the buttocks and genital area (called the saddle area or perineum)
- Painful cramping in the muscles of the legs that occurs during physical activity but is relieved promptly by rest (claudication)

When to see a doctor

Although ED may diminish a man's quality of life, it is not itself a dangerous condition. However, ED may be a symptom of a serious medical disorder. Because numbness in the groin or leg can be a sign of spinal cord damage, men who suddenly develop such numbness should see a doctor right away. Men who have other warning signs should call their doctor and ask how soon they need to be seen and examined.

What the doctor does

Doctors first ask questions about the man's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause for ED and additional tests that may need to be done (see table [Common Causes and Features of Erectile Dysfunction](#))

Doctors ask about

- Drug (prescription, recreational, and illicit) and alcohol use
- Smoking history
- History of diabetes
- History of high blood pressure
- History of atherosclerosis

- History of surgery (for example, for prostate enlargement, prostate or rectal cancer, or blood vessel disorders)
- History of injury (for example, a broken pelvic bone or a back injury)
- Symptoms of disorders of the blood vessels (for example, pain in the calves when walking or coolness, numbness, or blue color of the feet)
- Symptoms of nerve disorders (for example, numbness, tingling, weakness, incontinence, or falling)
- Symptoms of hormonal disorders (for example, loss of sex drive, increased size of breasts, decreased size of testes, loss of body hair, tremor, changes in weight or appetite, or difficulty tolerating heat or cold)
- Symptoms of psychological disorders, particularly depression
- Satisfaction with sexual relationships
- Sexual dysfunction (for example, vaginal dryness or depression) in the man's partner

Even though men may be embarrassed to talk to their doctors about some of these subjects, the information is important in determining the cause of ED.

The physical examination focuses on the genitals and prostate, but doctors also look for signs of hormonal, nerve, and blood vessel disorders and examine the rectum.

The cause is sometimes clear from the history. For example, ED may occur soon after prostate surgery or beginning a new medication. One important clue is whether erections are present at night or on awakening. When erections are present, a physical cause is less likely than a psychological cause because physical causes typically inhibit erections at all times. Other factors that suggest a psychological cause are sudden development in a young healthy man, occurrence of symptoms only in certain situations, and resolution of ED without any treatment. Claudication or coolness or a blue color in the toes or feet may indicate a problem with the blood vessels such as [peripheral vascular disease](#) or vascular disease caused by [diabetes](#).



Common Causes and Features of Erectile Dysfunction

Cause	Common Features*	Tests†
Blood vessel disorders		
Blockage of arteries (peripheral vascular disease)	Claudication (painful, aching, cramping, or tired feeling in the muscles of the legs that occurs regularly and predictably during physical activity but is relieved promptly by rest)	Comparison of blood pressures measured in the ankle and arm at the same time (called the ankle-brachial index)
	Usually risk factors (for example, high blood pressure, diabetes, or abnormal blood levels of cholesterol and lipids)	Testing for risk factors (for example, elevated blood glucose [sugar] and blood lipid levels) Ultrasonographic measurement of blood flow in the arteries of the penis
Venous leak (when the veins in the penis cannot prevent blood from leaving the penis during an erection, as they normally do)	Erections that occur but cannot be sustained	Ultrasonographic testing of the blood vessels of the penis
Nerve disorders		
Nerve damage caused by diabetes (diabetic neuropathy)	Known diabetes	A doctor's examination
	Sometimes numbness, burning, or other pains of the feet Sometimes urinary incontinence	Sometimes electromyography and nerve conduction studies
Multiple sclerosis	Intermittent episodes of weakness or numbness in different	MRI of brain and spinal cord



MSD
Nerve damage caused by diabetes (diabetic neuropathy)

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